Analytical Chemistry Cumulative Examination
December 5, 2014

1. A recent paper entitled " New Structural Insights into Mechanically Interlinked Polymers Revealed by Ion Mobility Mass Spectrometry" was published in *J. Amer. Chem. Soc.*, **134**, pp. 9193-9198 (2012). Please describe the specific type of polymer the research concerns. What are the main observations that ion mobility mass spectrometry revealed? This study involved use of a traveling wave ion mobility system. What is traveling wave ion mobility? How do the cross sections derived from those measurements differ from non-traveling wave ion mobility cross sections and can the two be related?

2. A review paper by Benesch, et. al. addresses the investigation of protein complexes in the gas phase (*Chem. Rev.* **107**, pp. 3544-3567 (2007)). Please explain how ion mobility mass spectrometry is used in this context. Name one such protein complex that was studied by this technique and explain what information was derived from those studies.

3. An interesting study of fragmentation and isomerization due to field heating in traveling wave ion mobility mass spectrometry was reported in a paper published in the August, 2014 issue of *J. Am. Soc. Mass Spectrom.*, (**25,** pp. 1384-1393) by Morsa, et. al.. Why is heating observed? What were some of the analytes employed in this study? How much heating was seen? What is the importance of this effect for large biomolecules? Why?

Biochemistry Cumulative Examination
12/05/2014

In your structures drawing for this exam you may use Fischer, Haworth, a plain old skeletal structure, or whatever style you prefer, and switch back and forth as you wish, but in all your drawings the stereochemistry must be clear.

1. What sugar(s) make(s) up sucrose? 10 points

2. Draw the structure of glucose. 11 points. Is the common form D or L? 3 points. Which did you draw? 1 point.

3. Draw the structure of N-acetylglucosamine. 15 points

4. Where are some places one commonly finds N-acetylglucosamine? 15 points

5. What sugar is found in DNA? Name it. 3 points. Draw the structure. 12 points

6. What are the two simplest (smallest) sugars typically commonly found in cells (albeit usually phosphorylated)? Name and draw each in unphosphorylated form. 20 points

7. What is the name we use for the carbon in a sugar that often has two oxygens bonded to it? 10 points